

Designing Workforce Management Systems for Industry 4.0

This book brings insight to the Human Resource (HR) management system and offers data-centric approaches and AI-enabled applications for the design and implementation strategies used for workforce development and management.

Designing Workforce Management Systems for Industry 4.0: Data-Centric and AI-Enabled Approaches focuses on the mechanisms of proposing solutions along with architectural concepts, design principles, smart solutions, and intelligent predictions with visualization simulation. Data visualization for the metrics of management systems and robotic process automation applications and tools are also offered.

This book is also useful as a reference for those involved in AI-enabled applications, data analytics, data visualization, as well as systems engineering and systems designing.



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>

Designing Workforce Management Systems for Industry 4.0

Data-Centric and AI-Enabled Approaches

Edited by
Alex Khang, Sita Rani, Rashmi Gujrati,
Hayri Uygun, and Shashi Kant Gupta



CRC Press

Taylor & Francis Group
Boca Raton London New York

CRC Press is an imprint of the
Taylor & Francis Group, an **informa** business

Designed cover image: © Shutterstock

First edition published 2024

by CRC Press

2385 Executive Center Drive, Suite 320, Boca Raton, FL 33431

and by CRC Press

4 Park Square, Milton Park, Abingdon, Oxon, OX14 4RN

CRC Press is an imprint of Taylor & Francis Group, LLC

© 2024 selection and editorial matter, Alex Khang, Sita Rani, Rashmi Gujrati, Hayri Uygün, and Shashi Kant Gupta; individual chapters, the contributors

Reasonable efforts have been made to publish reliable data and information, but the author and publisher cannot assume responsibility for the validity of all materials or the consequences of their use. The authors and publishers have attempted to trace the copyright holders of all material reproduced in this publication and apologize to copyright holders if permission to publish in this form has not been obtained. If any copyright material has not been acknowledged please write and let us know so we may rectify in any future reprint.

Except as permitted under U.S. Copyright Law, no part of this book may be reprinted, reproduced, transmitted, or utilized in any form by any electronic, mechanical, or other means, now known or hereafter invented, including photocopying, microfilming, and recording, or in any information storage or retrieval system, without written permission from the publishers.

For permission to photocopy or use material electronically from this work, access www.copyright.com or contact the Copyright Clearance Center, Inc. (CCC), 222 Rosewood Drive, Danvers, MA 01923, 978-750-8400. For works that are not available on CCC please contact mpkbookspermissions@tandf.co.uk

Trademark notice: Product or corporate names may be trademarks or registered trademarks and are used only for identification and explanation without intent to infringe.

Library of Congress Cataloging-in-Publication Data

Names: Khang, Alex, editor. | Rani, Sita, editor. | Gujrati, Rashmi, editor.

Title: Designing workforce management systems for industry 4.0 : data-centric and AI-enabled approaches / edited by Alex Khang, Sita Rani, Rashmi Gujrati, Hayri Uygün, Shashi Kant Gupta.

Description: First Edition. | Boca Raton : Taylor & Francis, 2024. | Includes bibliographical references and index.

Identifiers: LCCN 2023006447 (print) | LCCN 2023006448 (ebook) |

ISBN 9781032408248 (hardback) | ISBN 9781032412658 (paperback) |

ISBN 9781003357070 (ebook)

Subjects: LCSH: Personnel management--Technological innovations. |

Artificial intelligence. | Management--Data processing.

Classification: LCC HF5549.5.T33 D47 2024 (print) | LCC HF5549.5.T33 (ebook) |

DDC 352.6--dc23/eng/20230303

LC record available at <https://lcn.loc.gov/2023006447>

LC ebook record available at <https://lcn.loc.gov/2023006448>

ISBN: 978-1-032-40824-8 (hbk)

ISBN: 978-1-032-41265-8 (pbk)

ISBN: 978-1-003-35707-0 (ebk)

DOI: [10.1201/9781003357070](https://doi.org/10.1201/9781003357070)

Typeset in Times LT Std

by KnowledgeWorks Global Ltd.

Contents

List of Figures and Tables	ix
Preface.....	xv
Acknowledgments.....	xvii
Biography of Editors.....	xix
List of Contributors.....	xxi

Chapter 1	Workforce Management System: Concepts, Definitions, Principles, and Implementation.....	1
------------------	---	---

Snehal Maheshkar, Babasaheb Jadhav, and Alex Khang

Chapter 2	Industry Revolution 4.0: Workforce Competency Models and Designs	11
------------------	--	----

Alex Khang, Babasaheb Jadhav, and Sphurti Birajdar

Chapter 3	AI Powered Workforce Management in Industry 4.0 Era.....	35
------------------	--	----

Lokpriya Gaikwad and Alex Khang

Chapter 4	AI-Based Competency Model and Design in the Workforce Development System	47
------------------	--	----

Parin Somani, Shashi Kant Gupta, Chandra Kumar Dixit, and Anchal Pathak

Chapter 5	Data: An Anchor for Decision-Making to Build the Future Workforce Management System	65
------------------	---	----

Samiksha Shukla, Aerica Rishiraj, and Yash Lucas

Chapter 6	Data Mining Processes and Decision-Making Models in the Personnel Management System.....	85
------------------	--	----

Shashi Kant Gupta, Alex Khang, Parin Somani, Chandra Kumar Dixit, and Anchal Pathak

Chapter 7	Data-Driven Application of Human Capital Management Databases, Big Data, and Data Mining.....	105
------------------	---	-----

Alex Khang, Shashi Kant Gupta, Chandra Kumar Dixit, and Parin Somani

Chapter 8	Data-Centric Predictive Modeling of Turnover Rate and New Hire in Workforce Management System.....	121
	<i>Chandra Kumar Dixit, Parin Somani, Shashi Kant Gupta, and Anchal Pathak</i>	
Chapter 9	Impact of Artificial Intelligence (AI) on Talent Management (TM): A Futuristic Overview	139
	<i>Arpita Nayak, Ipseeta Satpathy, B.C.M. Patnaik, Sukanta Kumar Baral, and Alex Khang</i>	
Chapter 10	Data-Driven Artificial Intelligence (AI) Models in the Workforce Development Planning	159
	<i>Pallavi Jain, Vandana Tripathi, Ravisankar Malladi, and Alex Khang</i>	
Chapter 11	Prediction of Employees' Performance Using Machine Learning (ML) Techniques	177
	<i>Anchal Pathak, Chandra Kumar Dixit, Parin Somani, and Shashi Kant Gupta</i>	
Chapter 12	AI-Enabled Approaches and Models for Designing the Workforce through Training Systems for Physically Challenged People.....	197
	<i>Babasaheb Jadhav, Ashish Kulkarni, and Pooja Kulkarni</i>	
Chapter 13	Relevance Analytics of Work Motivation and Job Satisfaction in the Era of Industry 4.0.....	207
	<i>Mohammad Qais Rezvani, Hayri Uygun, Rashmi Gujrati, Shivangi Singh, and Nirmala Chaudhary</i>	
Chapter 14	A Bibliometric Analysis on Application of Artificial Intelligence (AI) in Workforce Management	217
	<i>Durga Prasad Singh Samanta, B.C.M. Patnaik, Ipseeta Satpathy, and Liji Panda</i>	
Chapter 15	Leveraging Employee Data to Optimize Overall Performance: Using Workforce Analytics	235
	<i>Saravanakrishnan V., Ginu George, Anson K. Joy, and Anusha B.</i>	

Chapter 16	Robotic Process Automation (RPA) Applications and Tools for the Workforce Management System	251
	<i>Sathya V., Jayashree K., and Malathi S.</i>	
Chapter 17	Exploring the Concept of Managing Women Employees' Work-Life Balance in Information Technology Company	265
	<i>Jansi Rani T. and Krishan Priya V.</i>	
Chapter 18	Challenges Faced by Marketers in Developing and Managing Contents in Workforce Development System	285
	<i>Ashwini Y. Sonawane and Alex Khang</i>	
Chapter 19	Linkages between Critical Success Factors and Factors of Workforce Performance in Remanufacturing Industry	311
	<i>Lalit K. Toke, Amol D. Lokhande, and Vishal S. Fegade</i>	
Chapter 20	A Study on the Impact of the Industry 4.0 on the Employees Performance in Banking Sector	331
	<i>Pooja Aggarwal and Alex Khang</i>	
Index		345



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>

List of Figures and Tables

Figure 1.1	The foundation of Industry 4.0. (Source: Flynn, Dance, & Schaefer, 2017.)	2
Figure 1.2	The thriving in terms of workplace diversity in the HR industry.	4
Figure 1.3	The four industrial revolutions.	5
Figure 1.4	The four worlds in global workforce.	6
Figure 2.1	Classification of competencies.....	17
Figure 2.2	Competency model.	17
Figure 2.3	An AI-oriented competency model (AIoCF) designed by SCEDEX organization, 2011.	19
Figure 2.4	Core competencies of workforce.	20
Figure 2.5	Core competencies of the workforce.	21
Figure 2.6	BUILD-UP competency framework designed by SCEDEX organization, 2011 (SCEDEX, 2011).	28
Figure 2.7	A BUILD-UP competency-based job description designed by SCEDEX organization. An example of a software engineer role (SCEDEX, 2011).	29
Figure 2.8	BUY-IN competency framework designed by SCEDEX organization, 2011.....	30
Figure 2.9	BUY-IN competency-based job description designed by SCEDEX organization, with the example of a software engineer role.	31
Figure 3.1	Illustration of recruitment workflow in the real world.	40
Figure 4.1	General framework of HCM.....	49
Figure 4.2	Flowchart of the proposed method.	52
Figure 4.3	Competency-based recruitment.	53
Figure 4.4	AI-based recruitment.....	55
Figure 4.5	Accuracy of the existing and proposed model.	57
Figure 4.6	The hiring time of the existing and proposed model.	58
Figure 4.7	Selection of skilled employees in the existing and proposed model.	59

Figure 4.8	Manager feedback on an existing and proposed model.....	60
Figure 4.9	Design principles of the competency model.....	61
Figure 5.1	Proposed framework for data-driven workforce identification.	69
Figure 6.1	A human resource management (HRM) system.	87
Figure 6.2	Schematic representation of the suggested optimization.....	89
Figure 6.3	Four steps in the data preprocessing lifecycle.	91
Figure 6.4	Genetic algorithm workflow.	93
Figure 6.5	A neural network.	94
Figure 6.6	A deep neural network.....	95
Figure 6.7	The WFL-DNN method's architecture.	98
Figure 6.8	Comparison of accuracy for existing and proposed methodologies (Bhoi et al., 2020; Protalinsky et al., 2019; Sheikh et al., 2019; Zhang et al., 2019).	99
Figure 6.9	Comparison of precision for existing and proposed methodologies (Bhoi et al., 2020; Protalinsky et al., 2019; Sheikh et al., 2019; Zhang et al., 2019).	100
Figure 6.10	Comparison of recall for existing and proposed methodologies (Bhoi et al., 2020; Protalinsky et al., 2019; Sheikh et al., 2019; Zhang et al., 2019).	101
Figure 6.11	Comparison of decision-making skills for existing and proposed methodologies.	101
Figure 6.12	Comparison of execution time for existing and proposed methodologies.	102
Figure 7.1	Application of big data in HRM.....	107
Figure 7.2	Flow of the proposed method.	110
Figure 7.3	Architecture of BCS-CNN.	113
Figure 7.4	Classification nodes of BCS-CNN.	114
Figure 7.5	Classification accuracy of the existing and proposed model.....	116
Figure 7.6	Precision of the existing and proposed model.....	117
Figure 7.7	Recall the existing and proposed model.....	118
Figure 7.8	F1-score of the existing and proposed model.	118
Figure 8.1	Factors of employee turnover.....	122
Figure 8.2	Proposed work of this research.	124

Figure 8.3	Schematic showing the bootstrap resampling technique.....	129
Figure 8.4	A representation of the structure of the EGB-RFA.....	130
Figure 8.5	Accuracy for proposed and existing methods.	131
Figure 8.6	Prediction error for proposed and existing methods.	132
Figure 8.7	Recall for proposed and existing methods.	133
Figure 8.8	F-measure for proposed and existing methods.....	133
Figure 8.9	Sensitivity for proposed and existing methods.....	134
Figure 9.1	Illustration of different main stages and benefits of AI.....	142
Figure 9.2	Study of the impact of AI on talent management.....	143
Figure 9.3	Illustration of AI in screening of curriculum vitae.	148
Figure 10.1	Three workforce planning categories.....	160
Figure 10.2	Workforce planning based on a data-driven AI model.	164
Figure 10.3	Importance of workforce planning.....	165
Figure 10.4	The process for workforce planning.....	168
Figure 11.1	Biases in the traditional employee performance evaluation system.	179
Figure 11.2	Framework of the proposed work.....	180
Figure 11.3	Flowchart of RF-based employee evaluation feature importance computation.....	185
Figure 11.4	Employee classification based on their performance using a WM-SVM.....	188
Figure 11.5	Comparison of employee performance classification accuracy of different models.	189
Figure 11.6	Comparison of employee performance classification precision of different models.	189
Figure 11.7	Comparison of employee performance classification precision of different models.	190
Figure 11.8	Comparison of time consumption for employee performance classification using different models.....	191
Figure 11.9	Number of employees versus their performance class.	192
Figure 12.1	Block diagram of an artificial intelligence-based online teaching system.....	200
Figure 12.2	Working method of the artificial intelligence-based online teaching system.....	202

Figure 14.1	The flow of the study is represented in graphic format	221
Figure 14.2	Visualization of clusters that represents the link strength between the coauthors with regard to co-occurrence and relativity in research.	223
Figure 14.3	Radar chart of the documents of the top ten authors.....	223
Figure 14.4	Co-occurrence author keywords network.....	224
Figure 14.5	Co-occurrence author keywords network in the areas of healthcare and education.	225
Figure 14.6	Co-occurrence author keywords network in the field of medicine and education.	226
Figure 14.7	Co-occurrence author keywords network.....	226
Figure 14.8	Documents by countries.	227
Figure 14.9	Documents by countries.	228
Figure 14.10	Documents by countries grouped by link strength.....	229
Figure 14.11	Visualization of organizations.	230
Figure 15.1	Relationship of attrition numbers.	245
Figure 15.2	Relationship of attrition numbers based on job satisfaction, age, and marital status.	245
Figure 15.3	Attrition based on job role.	246
Figure 16.1	Stages in RPA deployment.	256
Figure 16.2	Application of RPA applications.	258
Figure 16.3	RPA heat map for various financial applications.	259
Figure 16.4	RPA heat map for the banking industry.	259
Figure 17.1	Work-life balance of IT women employees.	267
Figure 17.2	Research methodology structure for work-life balance of IT women employees.....	274
Figure 17.3	Regression coefficients of the charts for work-life balance of IT women employees.....	277
Figure 19.1	The path of remanufacturing process.	316
Figure 19.2	Total variance explained for input factors.	322
Figure 19.3	Varimax-rotated factor matrix values for critical success factors.	323
Figure 19.4	Total variance for workforce performance measures.....	325

Figure 19.5 Varimax-rotated factor matrix values for workforce performance measures..... 326

Figure 20.1 Research model..... 335

Figure 20.2 Research model with obtained correlations..... 339

Table 2.1 Evolution of Workforce Competencies with the Industrial Revolution..... 14

Table 2.2 Definitions of Competency and Competency Model 14

Table 2.3 Competencies Necessary for the Workforce in Industry 4.0..... 22

Table 2.4 Competencies Necessary for the Workforce in the Digital Economy..... 22

Table 2.5 Transversal or Cross-Sector Competencies..... 23

Table 2.6 Essential Competencies for Various Domains of Industry 4.0 25

Table 2.7 AI and Data Science Literacy and Skills 28

Table 6.1 Representation of attributes..... 90

Table 6.2 Dataset Description 90

Table 8.1 Respective Ranges..... 125

Table 11.1 Demographic Characteristics of Sample Employees 181

Table 13.1 Several industry readiness and maturity models 208

Table 13.2 Work Motivation..... 210

Table 13.3 Job Satisfaction Measurement..... 212

Table 17.1 The Questions for Each Variable with the Topic 273

Table 17.2 Reliability Analysis 274

Table 17.3 Multiple Correlations..... 275

Table 17.4 Multiple Regressions 276

Table 17.5 The Level of Significance..... 276

Table 17.6 The Level of Significance..... 276

Table 19.1 Identified Critical Factors for Remanufacturing 317

Table 19.2 Identified Workforce Performance Measuring Factors for Remanufacturing..... 318

Table 19.3 Varimax-Rotated Factor Matrix on Sampled Data (Remanufacturing Factors)..... 323

Table 19.4 Varimax-Rotated Factor Matrix on Sampled Data
(Remanufacturing Workforce Performance Measures) 325

Table 20.1 The Demographic Profiles of the Respondents (N 469) 336

Table 20.2 Reliability and Validity Analysis 337

Table 20.3 Correlations among All Variables of the Research
Instrument 338

Table 20.4 Correlations between Industry 4.0 and Employees’
Performance 338

Table 20.5 Regressions among Five Variables of Industry 4.0 and
Employees’ Performance..... 340

Table 20.6 Hypotheses along with Their Statistical Indicators of
Relevance 340

Table 20.7 Regressions between Industry 4.0 and Employees’
Performance 341

Preface

In the era of the Fourth Industrial Revolution, we observe several powerful data-centric approaches, Internet of Things (IoT) technologies, cloud-based services, artificial intelligence (AI) science, machine learning (ML) models, deep learning (DL) frameworks, and data analytics used to support human capital management systems. Therefore, people are eager to leverage the most multifaceted data-centric approaches, AI-enabled applications, and emerging technologies to develop, deploy, and deliver next-generation workforce management systems (WMS).

Nowadays, resource-intensive competency models, methodologies, principles, tools, software, and networks are spreading across a variety of platforms and extensive supporting with the power of AI science and Data engineering. More precisely, the convergence of data-centric approaches, AI-enabled applications, and modern competency models are being touted as a conflux of next-generation solutions to design and implement a more flexible, and smarter human capital management system in order to deliver all kinds of connected as well as more intelligent services for all business operations.

Despite there are particular challenges associated with the adoption of data-centric approaches and AI-enabled applications in the complex operations of the WMS, and if they are integrated with full functionality to support the decision-making process, then we need these models and processes to articulate how they arrive at and process these decisions, as any wrong decision may result in irreparable damage. Hence, the trust and transparency of data-centric AI solutions are being seen as a critical challenge. Managers and business owners must insist on the unambiguous interpretability and explain the exact ability of AI-enabled systems' decisions. This is how the new disciplines of AI technology has flourished and is being recommended as a viable approach for overcoming this trust issue in workforce development and management.

The book brings insight into the WMS and offers data-centric approaches and AI-enabled applications for implementing the strategies of workforce development and management. It includes current developments, future directions, and also covers the concept of human capital management systems along with their sub-systems. It presents the insights of job performance analysis, building an efficient workforce, employee trainings, and many other concepts. It also focuses on the mechanisms of proposing solutions along with architectural concepts, design principles, smart solutions, and intelligent predictions with visualization simulation for the modern WMS.

The 20 chapters of this book are useful to researchers involved in AI-enabled applications, data analytics, and data visualization in the modern WMS. The book is also suitable as supporting material for undergraduate and graduate-level courses in related business administration disciplines.

Happy reading!

*Alex Khang, Sita Rani, Rashmi Gujrati,
Hayri Uygun, and Shashi Kant Gupta*



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>

Acknowledgments

This book is based on the concepts and design of data-centric, AI-enabled workforce management systems for the Industry 4.0 ecosystem. The idea to present a book to the readers in this domain was the brainchild of the editorial team. But it was not possible to bring it to reality without the hard work and trust of worthy contributors. Their effort and experience are rewarding for the academic world. So we especially want to acknowledge every individual who contributed to making this idea a success.

To all the reviewers we have had the opportunity to collaborate with and watch their hard work from afar, we acknowledge their tremendous support and valuable comments. Their support will also play a big role in the inspiration and foundation for future projects.

We also express our deep gratitude for all the advice, support, motivation, sharing, and inspiration we received from our faculty and academic colleagues.

And last but not least, we are grateful to our publisher, CRC Press (Taylor & Francis Group), for the wonderful support ensuring the timely processing of the manuscript and bringing this book to the readers.

Thank you, everyone.

Editorial Team: Alex Khang, Sita Rani, Rashmi Gujrati, Hayri Uygun, and Shashi Kant Gupta



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>

Biography of Editors

Dr. Alex Khang is a professor in information technology; AI and data scientist; software industry expert; and the chief technology officer (AI and Data Science Research Center) at the Global Research Institute of Technology and Engineering, North Carolina, United States. He has more than 28 years of teaching and research experience in information technology (software development, database technology, AI engineering, data engineering, data science, data analytics, IoT-based technologies, and cloud computing) at the Universities of Science and Technology in Vietnam, India, and the United States. He has been the chair session for 20 conferences, keynote speaker for more than 25 international conclaves, an expert tech speaker for 100 over seminars and webinars, an international technical board member for 10 international organizations, an editorial board member for more than 5 ISSNs, an international reviewer and evaluator for more than 100 journal papers, and an international examiner and evaluator for more than 15 Ph.D. theses in the computer science field. He has contributed to various research activities in fields of AI and data science while publishing many international articles in renowned journals and conference proceedings. He has published 52 authored books in computer science between 2000–2010; two authored books in software development; and 10 book chapters. He has published 10 edited books, and 10 edited books (calling for book chapters) in the fields of AI ecosystems (AI, ML, DL, robotics, data science, big data, and IoT), smart city ecosystems, healthcare ecosystems, fintech technology, and blockchain technology since 2020. He has over 28 years of nonstop work as a software product manager, data engineer, AI engineer, cloud computing architect, solution architect, software architect, database expert in foreign corporations of Germany, Sweden, the United States, Singapore, and multinationals (former CEO, former CTO, former Engineering Director, Product Manager, and Senior Software Production Consultant).

Dr. Sita Rani is an assistant professor, Department of Computer Science and Engineering, Guru Nanak Dev Engineering College, Ludhiana, Punjab, India. She completed her B.Tech and M.Tech degrees in the faculty of Computer Science and Engineering from Guru Nanak Dev Engineering College, Ludhiana. She obtained her Ph.D. in computer science and engineering from I.K. Gujral Punjab Technical University, Kapurthala, Punjab in 2018. She is an active member of IEEE and IAEngg. She is the receiver of ISTE Section Best Teacher Award and International Young Scientist Award. She has contributed to various research activities while publishing articles in the renowned journals and conference proceedings. She has published three international patents as well. Dr. Rani has delivered many expert talks in AICTE-sponsored faculty development programs and organized many international conferences during her 18 years of teaching experience. She is the member of editorial board for four reputed international journals.

Dr. Rashmi Gujrati is a professor, campus director, and dean of international affairs at Kamal Gandhi Memorial Ayurvedic College and Hospital, KC Group of Institutions, Nawanshahar, Punjab, India. She has authored ten books and eight book chapters, and published over 150 research papers in international journals. She is editor-in-chief of the *International Journal of Business Operation*, editorial board member of IJEFMS Science Publishing Group, USA, editorial board member of academia for the Global Business Advancement Association, USA, a reviewer for the Common Ground Research Networks at the University of Illinois, USA, editorial board member of the Tradepreneur Academic Platform at Southampton UK, and editorial board member of the *Journal of Agriculture, Food, Environment and Animal Sciences*, Turkey. She has attended over 200 national and international workshops, She has been a reviewer and keynote speaker at over 230 conferences.

Dr. Hayri Uygun, Recep Tayyip Erdogan University, Institute of Social Sciences, Business Administration, Turkey. He is editorial and advisory board member for *Tecnia Journal of Management Sciences* and *KC Journal of Engineering Management and Sciences*. He is also editor and vice president, Tradepreneur Global Academic Platform, UK; International Journal of Innovations In Business and Social Sciences Zien Journals.com. He is member of Digital Entrepreneurship ISBN No 978-93-5416-918-2; Covid 19 Impact of Foreign Direct Investment and Various Sector ISBN No 978-93-5408-401-0; Sustainable Development Goals-17 ISBN No:978-93-5473-218-8; Economics Renaissance ISBN NO: 978-93-5473-014-6; New Horizontal in Marketing Management ISBN 978-93-5437-081-6; Emerging New World ISBN: ISBN: 978-93-5445-402-8; Role of Technology in New Decade ISBN: 978-93-5473-307-9. He has published 40 papers in international journals and 12 papers in international conferences.

Dr. Shashi Kant Gupta, Ph.D., is a researcher and director at CREP, Lucknow, Uttar Pradesh, India. He completed his Ph.D. in computer science and engineering from Integral University, Lucknow, Uttar Pradesh, India. He worked as assistant professor in the department of computer science and engineering, PSIT, Kanpur, Uttar Pradesh, India. He has published more than 17 Indian patents; one patent is under grant approval. He has already been granted some German patents as well; He has more than 10 years of teaching experience and two years of industrial experience. His main research focuses on performance enhancement through cloud computing, big data analytics, IoT, and computational intelligence-based education. He is currently a reviewer for various international journals. He has published many Indian and Australian patents in the fields of information technology, computer science, and management.

List of Contributors

Pooja Aggarwal

School of Management, Babu Banarasi
Das University
Lucknow, Uttar Pradesh, India

Anusha B.

Department of Commerce (PG),
Krupanidhi Degree College
Bengaluru, Karnataka, India

Sukanta Kumar Baral

Department of Commerce
Indira Gandhi National Tribal
University (A Central University)
Amarkantak, Madhya Pradesh, India

Sphurti Birajdar

Global Business School and Research
Centre
Dr. D. Y. Patil Vidyapeeth (Deemed to
be University)
Pune, Maharashtra, India

Nirmala Chaudhary

University School of Management,
Kurukshetra University
Kurukshetra, Haryana, India

Chandra Kumar Dixit

Department of Physics
Dr. Shakuntala Misra National
Rehabilitation University
Lucknow, Uttar Pradesh, India

Vishal S. Fegade

Mukesh Patel School Of Technology
Management and Engineering
Shirpur, Maharashtra, India

Lokpriya Gaikwad

Department of Mechanical
Engineering
SIES Graduate School of Technology
Navi Mumbai, Maharashtra, India

Ginu George

Department of Commerce
CHRIST (Deemed to be University)
Bengaluru, Karnataka, India

Rashmi Gujrati

Campus Director, KC Group of
Institutions
Nawanshahr, Punjab, India

Shashi Kant Gupta

Integral University
Lucknow, Uttar Pradesh, India
CREP Pvt. Ltd.
Lucknow, Uttar Pradesh, India

Babasaheb Jadhav

Dr. D. Y. Patil Vidyapeeth (Deemed to
be University)
Global Business School & Research
Centre
Pune, Maharashtra, India

Pallavi Jain

Galgotias University
Greater Noida, Uttar Pradesh, India

Anson Kangirathingal Joy

Department of Commerce,
CHRIST (Deemed to be
University)
Bengaluru, Karnataka, India

Jayashree K.

Department of Artificial Intelligence
and Data Science
Panimalar Engineering College
Chennai, Tamil Nadu, India

Alex Khang

Universities of Science & Technology
Vietnam & United States
and
Global Research Institute of Technology
& Engineering
North Carolina

Ashish Kulkarni

Faculty of Management
MIT World Peace University
Pune, Maharashtra, India

Pooja Kulkarni

Faculty of Science & Technology
Vishwakarma University
Pune, Maharashtra, India

Amol D. Lokhande

Mechanical Engineering Department
Sandip Institute of Engineering and
Management
Nashik, Maharashtra, India

Yash Lucas

CHRIST (Deemed to be University)
Bengaluru, Karnataka, India

Snehal Maheshkar

Dr. D. Y. Patil Vidyapeeth (Deemed to
be University)
Global Business School & Research
Centre,
Pune, Maharashtra, India

Ravisankar Malladi

Department of CSE, GITAM
Hyderabad, Telangana, India

Arpita Nayak

KIIT School of Management
KIIT University
Bhubaneswar, Odisha, India

Liji Panda

KIIT School of Management
Kalinga Institute of Industrial
Technology (KIIT)
Bhubaneswar, Odisha, India

Anchal Pathak

Department of Management
College of Business & Economics
Bule Hora University
Bule Hora, Ethiopia

B. C. M. Patnaik

KIIT School of Management
Kalinga Institute of Industrial
Technology (KIIT)
Bhubaneswar, Odisha, India

Mohammad Qais Rezvani

Research Scholar
University School of Management
Kurukshetra University
Kurukshetra, Haryana, India

Aerica Rishiraj

CHRIST (Deemed to be University)
Bengaluru, Karnataka, India

Malathi S.

Department of Artificial Intelligence
and Data Science Panimalar
Engineering College
Chennai, Tamil Nadu, India

Durga Prasad Singh Samant

KIIT School of Management
Kalinga Institute of Industrial
Technology (KIIT)
Bhubaneswar, Odisha, India

Ipseeta Satpathy

KIIT School of Management
Kalinga Institute of Industrial
Technology (KIIT)
Bhubaneswar, Odisha, India

Samiksha Shukla

CHRIST (Deemed to be University)
Bengaluru, Karnataka, India

Shivangi Singh

University School of Management
Kurukshetra University
Kurukshetra, Haryana, India

Ashwini Y. Sonawane

Commerce and Management Faculty
Vishwakarma University
Pune, Maharashtra, India

Parin Somani

Department of Independent Academic
Scholar
London Organization of Skills
Development Ltd
London, UK

Jansi Rani T.

School of Management Studies
Vels Institute of Science Technology
and Advanced Studies
Chennai, Tamil Nadu, India

Lalit K. Toke

Mechanical Engineering Department
Sandip Institute of Engineering and
Management
Nashik, India

Vandana Tripathi

Computer Science & Engineering,
BBDITM
Lucknow, Uttar Pradesh, India

Hayri Uygun

Ardesen Vocational School
Recep Tayyip Erdogan University
Rize, Turkey

Krishan Priya V.

School of Management Studies
Vels Institute of Science Technology
and Advanced Studies
Chennai, Tamil Nadu, India

Saravanakrishnan V.

Department of Commerce
CHRIST (Deemed to be University)
Bengaluru, Karnataka, India

Sathya V.

Department of Artificial Intelligence
and Data Science Panimalar
Engineering College
Chennai, Tamil Nadu, India



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>

1 Workforce Management System

Concepts, Definitions, Principles, and Implementation

*Snehal Maheshkar, Babasaheb Jadhav,
and Alex Khang*

CONTENTS

1.1	Introduction	2
1.2	Related Work	2
1.2.1	Industry 4.0.....	2
1.2.2	Workforce	2
1.2.3	Types of Workforce	3
1.2.4	Workforce Diversity.....	3
1.2.5	Benefits of Workforce Diversity	4
1.2.5.1	The Red World Is All about Speed and Innovation.....	5
1.2.5.2	The Blue World Is All about Extreme Talent	6
1.2.5.3	The Green World Is All about the Company's Caring	6
1.2.5.4	The Yellow World Is All about Humans Coming First	6
1.2.6	Building a Connected Workforce for Industry 4.0	6
1.2.6.1	Continue to Combat Knowledge Loss	7
1.2.6.2	Place a Priority on People, Processes, and Culture	7
1.2.6.3	Strengthening the Employee Value Proposition	7
1.2.6.4	Rethinking the Talent Paradigm, Not Just Attracting and Retaining Employees	7
1.2.6.5	Lead with Quality	7
1.2.6.6	Set Direction with Comparative Benchmarking.....	7
1.2.6.7	Elevate and Support through Coaching and Education	7
1.2.6.8	Conducting a Job Analysis to Understand How Work Is Currently Done	7
1.3	Conclusion	7
	References.....	8

1.1 INTRODUCTION

In the twenty-first century, change is happening more quickly. There is strong competition for the best talent in the corporate world. The abilities that employers are searching for in their employees are changing as a result of automation and machines replacing human work.

One of a business’s most valuable resources for success is its personnel. To achieve efficiency and profitability, businesses must employ the proper amount of staff members with the appropriate skills and duties.

There are numerous workforce management advisors, consultants, and solution providers who manage all the tasks required to keep a creative staff, signaling the beginning of innovation in the workforce with the help of AI. Understanding the workforce separately and collectively is essential for comprehending the workforce through automation.

1.2 RELATED WORK

1.2.1 INDUSTRY 4.0

The rising technologies shown in Figure 1.1 are the foundation of Industry 4.0, or smart manufacturing, which has new demands on the manufacturing workforce in the age of AI and automation. The continued advancement of technology will enable machines and computers to eventually do tasks better than people (Bonekamp & Sure, 2015).

1.2.2 WORKFORCE

According to Oxford Dictionary, “the definition of *workforce* is any individual who is actively seeking employment or who is available to do so in a certain region, company, or industry.” The labor group employed by a business is known as the workforce. It is the labor market in the workplace.

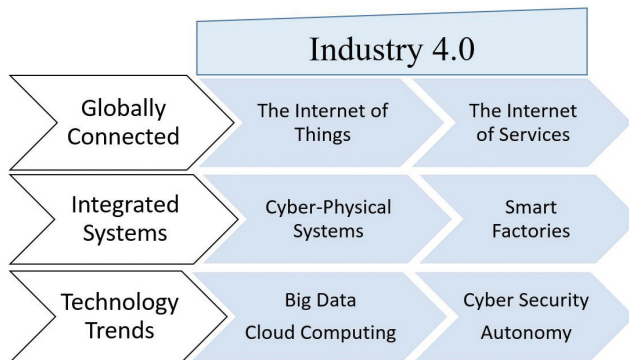


FIGURE 1.1 The foundation of Industry 4.0. (Source: Flynn, Dance, & Schaefer, 2017.)

The term *global workforce* refers to the international labor pool of workers, which may include immigrant workers, transient migrant workers, telecommuters, those in export-oriented employment, contingent workers, or other precarious employment, as well as those employed by multinational corporations, who are linked through a global system of networking and production.

A workforce is defined in the context of an organization as a collection of knowledge, experience, and abilities that people possess and apply to create value for the company (Agolla, 2018).

Industry 4.0 provides an opportunity to reinvent jobs and pay more attention to employees' skill development. The holistic method for managing human resources (HR) for Industry 4.0 lists four required employee competencies: technical, methodological, social, and personal (Khang et al., 2023).

Human capital will remain the key processor and decision-maker, which requires higher cognitive demands and mental tasks to control the developments that evolved from mechanization to computerization (Rana et al., 2021).

1.2.3 TYPES OF WORKFORCE

Employers may have as many as six or seven kinds of workers on board at any given moment in a labor market that has grown more complex. Seasonal and contingent workers work alongside full-time, part-time, and temporary employees. Here are different employee classifications employers may encounter.

- Full-time.
- Part-time.
- Temporary.
- Seasonal.
- Independent contractors.
- Freelancers.
- Temporary workers.
- Consultants.

1.2.4 WORKFORCE DIVERSITY

Due to growing globalization, interactions between people of different origins are necessary. People no longer live and work in isolated communities. The traditional workplace is evolving along with general workplace trends like more automation and digital technology, flexible working, and a desire for businesses to have a social purpose.

Corporations must prioritize utilizing and capitalizing on workplace diversity. *Workforce diversity* refers to policies and practices that aim to incorporate individuals within a workforce who are thought to be, in some manner, distinct from those in the dominant constituency.

Workforce diversity is the collective mixture of employees' differences and similarities, which includes characteristics, values, beliefs, experiences, backgrounds, preferences, and behaviors.

1.2.5 BENEFITS OF WORKFORCE DIVERSITY

More than ever, the human resources (HR) industry is thriving in terms of workplace diversity. People from all walks of life are employed by a diverse workforce. These are workers who will have various perspectives on the world and who will use their situations and experiences to offer fresh ideas that will help businesses to top the success iceberg. Researchers have listed a few concrete benefits of workplace diversity as [Figure 1.2](#).

The achievement of organizational goals, improved efficiency, better decision-making, growth, and development within an organization all depend on workplace diversity management and AI technologies. Workforce diversity management is an individual difference that can be visible (based on color, ethnicity, age, gender, physical ability, etc.) or invisible (based on education, competencies and skills, motivation, work experience, etc.) (Khanh & Khang, 2021).

So far, industrial revolutions include coal, gas, electronics, nuclear power, renewable energy, internet technology, and automation. From 1760 to the present, there has been an astonishing evolution. The entire terrain of the modern world has undergone numerous changes as a result of the different energy sources and, later, technologies that were discovered. The four industrial revolutions are briefly described as follows [Figure 1.3](#).

In Europe and North America, the First Industrial Revolution took place in the nineteenth century. During that time, rural, primarily agrarian communities transitioned to industrial, metropolitan societies. The agricultural community was impacted by the First Industrial Revolution.

The majority of the work was done manually. People who lost their farming livelihoods when factories occupied their former fields retrained to become boilermakers, ironsmiths, mechanics, etc. There was little demand for advanced skills (Bhambri et al., 2022).

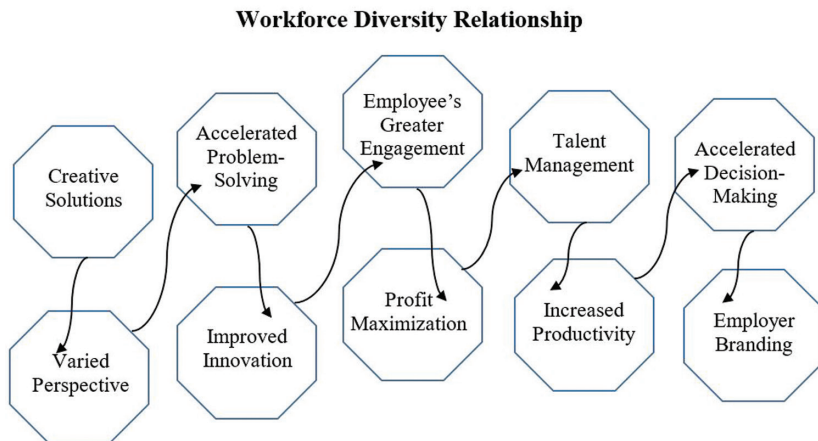


FIGURE 1.2 The thriving in terms of workplace diversity in the HR industry.

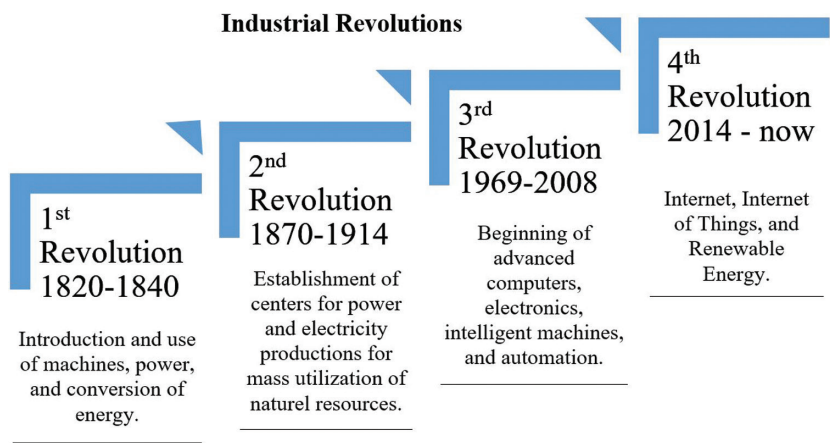


FIGURE 1.3 The four industrial revolutions.

The Second Industrial Revolution occurred just before World War I, between 1870 and 1914. It was a time of expansion for new industries as well as growth for those already established, including the mass manufacturing of steel, oil, and electricity.

The transition from analog electronic and mechanical equipment to digital technology is referred to as the Third Revolution, or Digital Revolution. The development of the PC, Internet, and ICT are examples of Third Industrial Revolution advancements. It was able to produce large quantities in tiny batches with a greater variety. Robots and computers have taken over many jobs (Hajimahmud et al., 2022).

High-speed internet, artificial intelligence, automation, big-data analytics, and cloud computing are four distinct technological advancements that are significantly responsible for the Fourth Industrial Revolution.

Looking at the current pace of innovation, Price Waterhouse and Coopers & Lybrand merge to create PricewaterhouseCoopers (PwC) believes that by 2030, the global workforce will be entirely different. It predicts that world employment could become one of four worlds: red, yellow, green, or blue as [Figure 1.4](#).

1.2.5.1 The Red World Is All about Speed and Innovation

The Red World is a perfect incubator for innovation in organizations and individuals what they want. Organizations and business owners compete to conceptualize and possess concepts that satisfy the needs of consumers. The greatest resource is human intelligence, and success is correlated with the demanded talent.

Performance in this context is more concerned with the outcome than the process. The Red World won't have a distinct HR department. Entrepreneurial leaders will rely on automated human processes and outsourced services. Larger organizations will search the globe for talent by utilizing AI and specialized talent strategists to find the experts they need (Tailor et al., 2023).